



Two stage stochastic bilevel programming model of a pre-established timberlands supply chain with biorefinery investment interests



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ABSTRACT

This work studies the supply allocation problem, using a Stackelberg game, for an established timberlands supply chain with an additional decision of new biorefinery investments. In a timberlands system, harvester and manufacturer decision makers have separate objectives to maximize their respective profits. This interaction is represented with a turn based Stackelberg game. The harvesters decide first on the quantity harvested, and the manufacturers decide on how much to utilize. This game is modeled with a bilevel mathematical program. The novel feature of this paper's bilevel formulation is the inclusion of parametric uncertainty in a two stage model. The first stage problem involves logistical decisions around biorefinery investments, such as location and capacity, while the second stage problem involves a bilevel timberlands model with parameter uncertainty. Studying this problem formulation revealed interesting insights for solving multiperiod problems with bilevel stages as well as the decision maker's behavior for the timberlands model.

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1. Introduction

Biofuel production is a way to replace traditional fossil fuels to meet a portion of demand for liquid fuels. Biofuels are a promising choice for sustainable transportation fuel due to the limited changes needed to existing fuel distribution and combustion technologies, particularly when converted to direct “drop-in” diesel or gasoline replacements. Timber industry management has shown interest in biofuels for market diversification. Timber demand varies due to dips in housing construction and experiences secular decline as new technologies replaces paper products, so timberland owners are looking for new outlets for timber materials. Biofuel production looks promising in this regard because it can utilize both harvested biomass and timber residuals discarded by other processes in the timberlands network. With enough residuals, significant amounts of biofuel can be produced and new revenue can be generated from them. However, investment in such a large scale project has many financial risks owing to many uncertainties. Supply chain modeling and analysis can be used to simulate adding

biorefineries to the existing timberlands system and clarify some aspects of the investment as well as reveal insights about decision making behavior.

There have been many recent studies involving biofuel supply chains. Sharma et al. (2013) developed a biorefinery model that determined stakeholder value as well as optimal production decisions. Elia et al. (2013) developed a large scale mixed integer linear programming (MILP) model that simulates hardwood biomass resource flows across the entire United States. Andersen et al. (2013) used an MILP model for the design and planning of an integrated ethanol and gasoline supply chain. Daoutidis et al. (2013) provided an in depth discussion on the status of biofuel research and future directions that could be pursued. Santibanez-Aguilar (2013) proposed utilizing the water hyacinth for biofuel production as a sustainable way to remove the harmful plant from water systems and developed a supply chain model to simulate this solution. Floudas et al. (2012) provided a more general review for studies of hybrid and single feedstock liquid fuels. The paper covered biomass processes and some discussion of optimization with uncertainty.

Biofuel supply chain modeling with game theory approaches has been studied in the past. Bai et al. (2012) utilized Stackelberg game theory to model resource providers and manufacturers for a biofuel supply chain as separate decision making entities. They performed case studies to determine how cooperative or non-cooperative behavior effected the decisions made in the system.

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Wang et al. (2013) also created a bilevel model with a Stackelberg game approach that gives insights on biofuel supply chain design under a government mandate for minimum biofuel production per year. These two papers focus on biofuels derived from food and specifically focus on the food versus fuel problem. In these studies, the interaction between the biorefinery and the biomass providers are studied with bilevel programming.

Our previous research also explored Stackelberg games and bilevel programming to model the biofuel problem, but our study does not involve biofuels synthesized from food type biomass (Yeh et al., 2014). Instead, we look at the biofuel production problem in a timberlands supply chain. We hypothesized that using this game theory approach would provide another perspective on the interactions between two major decision making sectors in the timber network: harvesters and manufacturers. Stackelberg games are a two player turn based game with a leader (first decision maker) and follower (second decision maker) (Bard, 1998). Both players have a separate set of objectives, constraints, and decisions, but have full knowledge of each other's information. The players are connected in that their decisions influence each other. As a single period turn based game, it is assumed that, after the leader makes his decision, the follower will react optimally to the given situation. Knowing this, and having the full information state, the leader can anticipate the behavior of the follower. Stackelberg games are most commonly represented with bilevel programming. Our initial study compared a single period bilevel model with a single period single level model. The results from previous analysis showed that representation in a bilevel form revealed decisions and behavior very different from the single level model (Yeh et al., 2014).

After this study on the viability of bilevel models with our timberlands system, we have continued to develop the model by including uncertainty. Investment decisions are important in supply chain modeling and must be made facing the unpredictability of macro-economic conditions and the uncertainty in technological parameters in the early-to-mid stage of process development. One of the earlier studies in biofuel supply chains under uncertainty was performed by Dal-Mas et al. (2011). Dal-Mas et al. (2011) developed an MILP model for the design of a corn to ethanol supply chain for a 10 year time horizon. This model determined optimal location logistics with product price and production cost uncertainty for maximizing profit and minimizing risk case studies. For the optimal design of hydrocarbon biorefinery supply chains, Gebreslassie et al. (2012) developed a multiperiod stochastic MILP model that included demand and supply uncertainties. Mansoornejad et al. (2013) designed a model with uncertainty in market scenarios and system network scenarios for a forest biorefinery supply chain. They used a stepwise methodology to determine the effect of design decisions on process operations. Sharma et al. (2013) also designed a biofuel production model with market uncertainty. The model covered strategic investment decisions for the biofuels through a stochastic integer programming model. Tong et al. (2014) looked at an integrated biofuel and petroleum supply chain system. The system was modeled stochastically as a two stage MILP with scenarios involving biomass availability, fuel demand, crude oil prices, and technology evolution. It was assumed that through technology evolution, material, production, and infrastructure costs would decrease.

Studies have also been performed with bilevel models and uncertainty. Capitanescu and Wehenkel (2013) determined the worst case scenarios for power flows under operational uncertainty for the determination of contingency plans. Their bilevel program was solved through heuristic approximation. A traffic control problem with traffic uncertainty was studied by Chiou (2014). A min-max bilevel program was utilized where the leader managed traffic signals and the followers were the drivers. Konur and

Golias (2013) tackled a truck scheduling problem with arrival time uncertainty. They used a genetic algorithm approach to determine optimal time assignments for pessimistic and optimistic situations. Wogrin et al. (2011) developed a model that calculates long term investments in the electricity market with regards to the investments of other companies. The investing company was modeled in the upper level while the competing company decisions were considered in the lower level. Competition uncertainty was modeled through pricing variations.

The possible introduction of biofuel production within an uncertain environment means the timing of the biorefinery investment decisions must be defined in the model. We assume the biorefinery investment decisions must be set before the uncertain scenarios occur. Therefore, the model can be represented as a two stage stochastic integer program (Sahinidis, 2004). In the first stage, the biorefinery binary decision will be set. The second stage will be a scenario based uncertainty bilevel model. Multiperiod problems with uncertainty have been used extensively in system modeling (Birge and Louveaux, 2011). Paules and Floudas (1992) developed a two stage stochastic mixed integer non-linear program (MINLP) to model a multiperiod heat-integrated distillation design problem. Uncertain parameters included feed composition and flowrate, and the uncertainty was represented through discrete scenarios. Kang and Lansey (2012) used a multiperiod model to study water supply infrastructure planning. They used discrete scenarios to represent uncertainties such as increased demand from population growth, changes in regulation, and public outlook. This model analyzed the situation to determine the timing of water infrastructure investments and their size. Moreno and Montagna (2012) developed a two stage stochastic multiperiod linear generalized disjunctive programming model to study the design and planning decisions of multiperiod batch facilities with consideration to demand uncertainties. The design decisions were made in the first stage before information about the product demand was known. Zhu et al. (2011) also studied batch plant production in a multiperiod sense but approached the system with a scheduling problem. They modeled the system with a two stage stochastic integer programming model for the production of multiple products with demand uncertainties. Their model included penalties for production shortfalls and excess. Rodriguez et al. (2014) considered an inventory management problem with demand uncertainty with an MINLP. The goal of this work was to propose redesign decisions for a spare parts supply chain. With the non-linear behavior in consideration, a piecewise linearization approach was used to determine the lower bound of the optimal solution. Giarola et al. (2013) developed a multiperiod MILP model to optimize an ethanol production supply chain with uncertain market conditions. The supply chain was studied over an 18 year horizon divided into 6 time periods. The model considered factors such as carbon cost emissions, biomass crop management, and technology learning issues. Three case studies were considered: optimal economic and environmental situation, risk mitigation under an optimal economic situation, and risk mitigation under an optimal environmental situation. Kim et al. (2011) developed a two stage mixed integer stochastic program around a timberlands supply chain with biofuel production. This study utilized discrete scenarios generated by uncertain parameters to determine the optimal estimated value of the system.

There is also research involving multiperiod bilevel games, but literature with this methodology is rarer. Sinha et al. (2014) modeled an oligopolistic market with a bilevel model over multiple time periods. This research studied and solved the general formulation of a multiperiod multiple-leader-follower system. Su and Geunes (2013) designed a two stage supply chain bilevel model with uncertainty. They studied a supplier pricing problem where

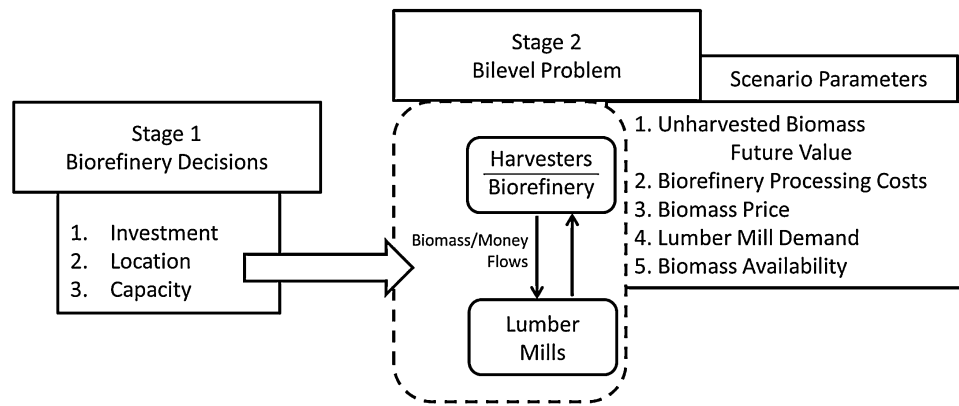


Fig. 1. Two stage model system diagram.

the supplier must set prices over a finite time horizon with demand uncertainties from the retailer. The bilevel model calculated the supplier's pricing decisions and the retailer's operation decisions.

This research extends the work of Yeh et al. (2014) by exploring bilevel problems in a multistage form for representing biofuel investment decisions. The single period model from the previous study is expanded to a multiperiod two stage model with parameter uncertainty. The first stage decisions concern biorefinery investment. The second stage is the bilevel supply allocation model representing the behavior of harvesters and manufacturers in the timberlands supply chain. The multiperiod format captures the timing aspect of the biorefinery investment decision. Adding uncertainty to specific parameters creates a more realistic model that will ultimately be more useful as an analysis tool.

This paper studies the timberlands system's decision making behavior for investment in a biorefinery under economically uncertain future scenarios and within the context of a second player in the supply chain that pursues its own objectives with regards to material utilization once the biorefinery decision has been made. The study is extended to examine the robustness of the decision to removal of different system assets, specifically lumber mills and harvestable lands. It is shown that the closure of lumber mills can negatively impact biorefinery decisions because the volume of lumber mill and forestry residuals to the refinery is reduced.

2. Problem statement

2.1. System overview

A bilevel game, formulated to model an existing timberlands system, is extended to a two stage stochastic program (Birge and Louveaux, 2011). First stage decisions involve the biorefinery investment, and the second stage decisions are the flows of material to the biorefineries and manufacturing facilities. Harvesters provide saw logs to lumber mills and produce logging residuals as byproducts. Mills convert logs into lumber boards and produce shavings as a byproduct. The biorefinery can utilize logs, logging residuals, and shavings to produce gasoline. The first stage decision is represented through a set of binary variables corresponding

to a combination of 3 locations and 3 capacities. For this study, a maximum of 1 biorefinery could be built, giving a total of 10 possible investment decisions: 9 decisions comprising of the different location/capacity combinations and the remaining decision being no investment. A diagram of the system is shown in Fig. 1.

The second stage model is a bilevel timberlands model with harvesters as the leaders and manufacturers as the followers. Multiple second stage scenarios are generated as periods, with each period housing a different set of uncertain parameter values, the standard stochastic programming problem. This orientation of harvesters and manufacturers were set based on discussions with our industrial partner. The harvesters are considered upper management, and thus, should be the ones deciding on the biorefinery implementation as the leaders of the bilevel system. Harvesters provide timber material for manufacturers to process. Both levels are composed of multiple entities within the level; however, each entity's goal is not to maximize their own profit but the level's profit as a whole. This is represented by the dashed border around the two problems in Fig. 1. Both levels maximize their own overall profit under different conditions. The manufacturer level has the short term goal of maximizing their present profit. In the leader level, the dynamic nature of timber resources requires the harvesters to plan over the long term. In this model, unharvested biomass is given a value that simulates potential future revenue. The harvester must find an optimal balance between harvested and unharvested resources. In this study, the system is comprised of 15 harvest zones providing logs to 4 lumber mills to produce timber.

It is assumed that the first stage biorefinery location and capacity decisions are controlled by the harvesters. This reflects a business case where the harvesters are seeking to diversify the portfolio of their outlets and maximize the efficiency of the existing biomass resources. This placement adds another set of decisions that the upper level must consider due to potential revenue from the biorefinery investment. Furthermore, the biorefinery facility creates additional dynamics in the system. Biorefineries can produce gasoline precursors from previously discarded residuals from harvesting and manufacturing processes, meaning new biomass types and resource flows are created. In a deterministic case, the first stage decisions could be combined with the harvester decisions

Table 1
Scenario parameter values.

Future value weight	Biorefinery processing cost multiplier	Biomass cost multiplier	Minimum processing demand	Biomass availability
w_{kj}	μ_j	δ_j	t_j	v_{ikj} and q_{ikhj}
0.7	0.7	1	0	Scen. 1 (Table A.10)
1.1	1	1.5	50% 90%	Scen. 2 (Table A.10)

Table 2
Indices.

i	Harvest source
m	Manufacturer sink
k	Biomass type
l	Product type
h	Harvest interval
u	Biorefinery capacity index
e	Biorefinery location index
j	Scenario index

Table 3
Scenario independent parameters.

c_{ik}	Selling price from harvester i for biomass type k
$\hat{c}_{mk}$	Selling price from manufacturer m for biomass type k
β_{ue}	Annualized capital cost of biorefinery of capacity u at location e
o_{ik}	Selling price of harvester i 's biomass type k to external buyers
b_{ik}	Lower bound of harvest decision x_{ik1}
f_{ik}	Fixed cost of harvesting biomass type k at harvester i
r_{ikh}	Processing cost per unit of biomass type k harvested at harvester i in harvest interval h
α_{kl}	Conversion factor for biomass type k into product l
s_{im}	Shipping cost from harvester i to manufacturer m
$\hat{s}_{ie}$	Shipping cost from harvester i to biorefinery at location index e
G_m	Maximum capacity of manufacturer m
$\hat{G}_u$	Biorefinery capacity for index u
A_{ml}	Maximum capacity of product l that can be produced from manufacturer m
p_{ml}	Selling price from manufacturer m of product l
$\hat{p}_{lue}$	Selling price of biorefinery at location e of capacity u of product l
a_k	Approximated average value of biomass type k for future value calculation
$\gamma_{kk'}$	Fraction of resources of biomass type k' obtained from harvesting biomass type k at the harvester
$\hat{\gamma}_{kk'}$	Fraction of resources of biomass type k' obtained from processing biomass type k at the manufacturer
g_{mke}	Price of biomass type k sold by manufacturer m to biorefinery at location index e
n_m	Processing cost per unit of material for manufacturer m
$\hat{n}_u$	Processing cost per unit of material for biorefinery of capacity index u
κ	Number of biorefineries that can be introduced to the system

and solved directly. However, for these more strategic decisions, parametric uncertainty may be incorporated using a multiperiod model. Each period represents a single scenario that contains a combination of uncertain parameters. The decision timeline is constructed as follows. First, the strategic decisions (the size and location of the biorefinery) are fixed using an expected value of the objective function obtained by the standard technique of weighting each scenario. However, the solution to each scenario is found by the bilevel representation of the game between harvester and manufacturer, as opposed to the more traditional set of constraints. In this study, we construct the simplex of scenarios based on taking each parameter value combination, a total of 48 scenarios (Table 1).

3. Model formulation

Tables 2–5 detail the indices, parameters, and decision variables of the timberlands supply chain model with a biorefinery. The parameter values are given in the appendix.

3.1. Constraints

Constraints (1) and (2) are the bounds on the harvester decision x_{ikhj} . The h index represents different levels of harvest available

Table 4
Scenario dependent parameters.

q_{ikhj}	Upper bound of harvested quantity from harvester i of biomass type k in harvest interval h for scenario j
v_{ikj}	Amount of harvestable resources in harvester i for biomass type k for scenario j
δ_j	Biomass pricing level for scenario j
μ_j	Biorefinery processing cost multiplier for scenario j
t_j	Lumber mill demand level modifier for scenario j
w_{kj}	Weighting of average value of biomass k for future value calculation for scenario j
ω_j	Occurrence probability weight for scenario j

to the timberlands managers. In this study, 2 levels exist, the preplanned level and the extra level. Harvesting from the extra level incurs an additional cost. The bounds are controlled by binary variable z_{ikj} , the activity decision of a harvester. The lower bound of an active harvester (Constraint (1)) is a start-up amount; this prevents the harvest of unrealistically small amounts of material. This lower bound only constrains the preplanned level of material ($h = 1$). In this study, b_{ik} has a value of 0.5, which, along with the values of the fixed costs of harvesting, were chosen to simulate an economy of scale. As the harvester provides more material, the relative importance of the fixed cost declines. The upper bound (Eq. (2)) is limited by the amount of available material in the scenario. The preplanned amount (q_{ikhj} when $h = 1$) is 75% of the available material (v_{ikj}), and the extra level (q_{ikhj} when $h = 2$) is the remaining 25%.

$$x_{ik1j} \geq b_{ik} z_{ikj} \quad \forall i, k, j \quad (1)$$

$$x_{ikhj} v_{ikj} \leq q_{ikhj} z_{ikj} \quad \forall i, k, h, j \quad (2)$$

Eqs. (3)–(6) are constraints on the biorefinery. Constraint (3) limits the number of biorefinery investments to κ . This study was limited to 1 biorefinery ($\kappa = 1$). Constraint (4) limits the number of biorefineries in a location to 1. Constraint (5) limits the amount of purchasable biomass by the amount of byproduct generated by the manufacturers. Byproduct generation of biomass type k from biomass k' is given by parameter $\hat{\gamma}_{k'k}$. In our study, $\hat{\gamma}_{k'k} = 0.3$ dry tons of shavings/wet ton of logs processed (Hanks, 1980). The upper limit on purchased material by the biorefinery is determined by Constraint (6). The capacity is determined by binary decision variable D_{ue} , which sets the location and size of the biorefinery. In this study, the gasoline demand on the biorefinery from external retailers is 0. This allows us to see the optimal operating conditions for

Table 5
Decision variables.

<i>First stage variables</i>	
D_{ue}	Binary decision of biorefinery capacity index u of at location e
<i>Second stage variables</i>	
z_{ikj}	Binary variable that controls the harvester i 's activity for harvesting biomass type k for scenario j
x_{ikhj}	Fraction of harvested resources versus available harvestable resources in harvester i for biomass type k at harvest interval h for scenario j
y_{imktj}	Purchase quantity from harvester i by manufacturer m of biomass type k for product l for scenario j
π_{ikluej}	Purchase quantity from harvester i of biomass type k for product l at location index e and capacity index u for scenario j
$\hat{\pi}_{mkluej}$	Purchase quantity from manufacturer m of biomass type k for product l at location index e and capacity index u for scenario j

biofuel production under the full decision making flexibility. This assumption is also seen in Yeh et al. (2014)

$$\sum_u \sum_e D_{ue} \leq \kappa \quad (3)$$

$$\sum_u D_{ue} \leq 1 \quad \forall e \quad (4)$$

$$\sum_l \sum_u \sum_e \hat{\pi}_{mkluej} \leq \sum_{k'} \hat{\gamma}_{k'k} \sum_i \sum_{l'} y_{imk'l'j} \quad \forall m, k, j \quad (5)$$

$$\sum_k \sum_l (\sum_i \pi_{ikluej} + \sum_m \hat{\pi}_{mkluej}) \leq D_{ue} \hat{G}_u \quad \forall u, e, j \quad (6)$$

3.1.1. Manufacturer

The manufacturer level is constrained by facility capacity (Constraint (7)), retailer demand (Constraint (8)), and material availability (Constraint (9)). Retailer demand is modified by scenario parameter t_j . In Constraint (9), material availability is dependent on harvester decision x_{ikhj} as well as logging residual yield parameter $\gamma_{k'k}$. For this study, $\gamma_{k'k} = 0.3$ dry tons of residuals/wet ton of logs harvested (Jurevics, 2010).

$$\sum_i \sum_k \sum_l y_{imklj} \leq G_m \quad \forall m, j \quad (7)$$

$$\sum_k \alpha_{kl} \sum_i y_{imklj} \geq t_j A_{ml} \quad \forall m, l, j \quad (8)$$

$$\sum_l (\sum_m y_{imklj} + \sum_u \sum_e \pi_{ikluej}) \leq v_{ikj} \sum_h x_{ikhj} + \sum_{k'} \gamma_{k'k} (v_{ik'j} \sum_{h'} x_{ik'h'j}) \quad \forall i, k, j \quad (9)$$

3.2. Objective function

The estimated value of the upper level objective function is found by summing over all scenarios with weighting. Each scenario is multiplied to a weight parameter (ω_j), which corresponds to the scenario's probability of occurrence. In this study, all 48 scenarios of parameter values are evenly weighted, giving an occurrence probability of 0.0208.

3.2.1. Upper level value

The value of the harvester is determined by the revenue from biomass sales balanced against the unharvested resource value. A future value is assigned to simulate timberland dynamics by discouraging the harvest of all available resources. In a realistic situation, the harvest managers would save material for future time periods. Giving unharvested material a value forces the upper level decision makers to balance the benefits of harvesting and not harvesting. The future value is calculated by weighting the average present value of the biomass. The weighting is given by parameter w_{kj} while the average value a_k is calculated by Eq. (13). In Eq. (13), i refers to the number of harvester entities in the upper level. The average value of the material is approximated to be 90% of the average internal price and 10% of the average external price. The internal (Eq. (10)) and external (Eq. (11)) revenue also contribute to the upper level value. Internal revenue is generated from biomass sales to the biorefinery and manufacturer. External revenue comes from leftover material that is sold to facilities external to the supply chain. We utilized a set pricing for the biomass instead of an equilibrium equation. An equilibrium based pricing utilizing

supply-demand interactions would require a non-linear representation. A non-linear pricing scheme would create a more difficult set of optimal KKT conditions to solve, discussed more in depth in Section 4. With an already complex formulation, we wanted to maintain a level of simplicity to allow for a realistic sized supply chain model that was solvable in a reasonable time. The upper level objective function also includes revenue from the biorefinery (Eq. (14)).

$$\text{Internal revenue for scenario } j = \sum_i \sum_k \delta_j c_{ik} \sum_l (\sum_m y_{imklj} + \sum_u \sum_e \pi_{ikluej}) \quad (10)$$

External revenue for scenario j

$$= \sum_i \sum_k o_{ik} (v_{ikj} \sum_h x_{ikhj} + \sum_{k'} \gamma_{k'k} v_{ik'j} \sum_{h'} x_{ik'h'j} - \sum_l (\sum_m y_{imklj} + \sum_u \sum_e \pi_{ikluej})) \quad (11)$$

Unharvested resource value for scenario j

$$= \sum_k w_{kj} a_k \sum_i v_{ikj} (1 - \sum_h x_{ikhj}) \quad (12)$$

$$a_k = 0.9i^{-1} \sum_i c_{ik} + 0.1i^{-1} \sum_i o_{ik} \quad (13)$$

Shared biorefinery revenue for scenario j

$$= \sum_l \sum_u \sum_e \hat{p}_{lue} \sum_k \alpha_{kl} (\sum_i \pi_{ikluej} + \sum_m \hat{\pi}_{mkluej}) \quad (14)$$

3.2.2. Manufacturer revenue

The manufacturer's objective function is not weighted with a probability. The manufacturer behaves by reacting to the decision of the harvester and finds the optimal solution for the situation, which is scenario dependent. Therefore, there exists an objective function for each scenario. The formulation below shows the revenue, costs, and value for each scenario j . The manufacturer's revenue is gained from product sales and byproduct sales to the biorefinery.

Manufacturer revenue for scenario j

$$= \sum_m \sum_l p_{ml} \sum_k \alpha_{kl} \sum_i y_{imklj} \quad (15)$$

$$\text{Byproduct revenue for scenario } j = \sum_m \sum_k \hat{c}_{mk} \sum_l \sum_u \sum_e \hat{\pi}_{mkluej} \quad (16)$$

3.2.3. Upper level costs

The only costs to harvesters are the processing costs (Eq. (17)). f_{ik} is the harvesting startup cost while r_{ikh} is the cost of harvesting from the extra level of material. r_{ikh} is 0 for $h = 1$. The biorefinery investments incur all other costs in the objective function. These costs include an annualized capital cost for the investment (Eq. (18)), the biomass acquisition costs from harvesters and manufacturers

(Eqs. (19) and (20)), transportation costs from harvesters (Eq. (21)), and processing costs (Eq. (22)). The acquisition costs of biomass from manufacturers (g_{mke} in Eq. (20)) is a combination of the price to purchase the biomass and the transportation costs.

$$\text{Processing costs for scenario } j = \sum_i \sum_k (f_{ik} z_{ikj} + v_{ikj} \sum_h r_{ikh} x_{ikhj}) \quad (17)$$

$$\text{Annualized capital cost for all scenarios} = \sum_u \sum_e \beta_{ue} D_{ue} \quad (18)$$

$$\begin{aligned} \text{Acquisition costs from harvester for scenario } j \\ = \sum_i \sum_k \delta_{c_{ikj}} \sum_l \left(\sum_u \sum_e \pi_{ikluej} \right) \end{aligned} \quad (19)$$

$$\begin{aligned} \text{Acquisition costs from manu. for scenario } j \\ = \sum_m \sum_k \sum_e g_{mke} \sum_l \sum_u \hat{\pi}_{mkluej} \end{aligned} \quad (20)$$

$$\begin{aligned} \text{Biorefinery transportation costs from harvesters for scenario } j \\ = \sum_i \sum_e \hat{s}_{ie} \sum_k \sum_l \sum_u \pi_{ikluej} \end{aligned} \quad (21)$$

$$\begin{aligned} \text{Biorefinery processing costs for scenario } j \\ = \mu_j \sum_u \hat{n}_u \sum_e \sum_k \sum_l \left(\sum_i \pi_{ikluej} + \sum_m \hat{\pi}_{mkluej} \right) \end{aligned} \quad (22)$$

3.2.4. Manufacturer costs

Similar to the biorefinery, the manufacturer costs involve acquisition (Eq. (23)), transportation (Eq. (24)), and processing costs (Eq. (25)).

$$\begin{aligned} \text{Acquisition costs from harvester for scenario } j \\ = \sum_i \sum_k \delta_{c_{ikj}} \sum_l \sum_m y_{imklj} \end{aligned} \quad (23)$$

$$\text{Transportation costs for scenario } j = \sum_i \sum_m s_{im} \sum_k \sum_l y_{imklj} \quad (24)$$

$$\text{Processing costs for scenario } j = \sum_m n_m \sum_i \sum_k \sum_l y_{imklj} \quad (25)$$

A few further assumptions are made regarding the system model. First, it is assumed that, by contractual obligation, sufficient material must be provided from the harvester so that the manufacturer can satisfy its minimum final demand requirements. This also ensures that the harvester cannot drive the problem to an infeasible state. Second, the demand constraint on manufacturers is only a lower bound. Any products manufactured in excess are sold at full price. Furthermore, all purchased materials are manufactured into product. Any harvested material left unpurchased is sold to buyers external to the system for a lower price.

4. Solution methods

The bilevel problem can be simplified by introducing the lower level optimal (KKT) conditions as constraints into the upper level, reducing the problem to a single level non-linear program (NLP). The bilinear constraints in the NLP are converted to linear constraints with binary variables and a big M term, forming an MILP. These methods are the same as in Yeh et al. (2014), and an example derivation is presented in that paper.

Table 6
Benchmark results for enumeration.

M values	Varied	200	300	400	500	600
Total calculation times (s)	45.15	57.85	70.82	69.71	70.60	88.43

This reformulation method of solving bilevel problems is commonly used (Bard, 1998). In this study, the fully converted model was solved in three ways: the first required solving the full multiperiod two stage model; the second enumerated the biorefinery decisions and compared the objective functions; the third enumerated through every scenario and biorefinery decision combination. In the latter two cases, the number of bilevel problems corresponds to the number of scenarios being studied and hence does not scale effectively as the number of binary decisions increases. In contrast, although the first method can use sophisticated branch-and-bound algorithms to avoid this enumeration, the algorithms have not been optimized for the structure of this problem type and hence, for smaller problems, are substantially slower.

5. Results

Optimization was performed in GAMS (version 23.5.1) using the CPLEX solver (version 12.2) on an Intel Core i7-3510QM at 2.3 GHz and 16 GB RAM. The lower level problem in the bilevel model was converted to KKT conditions with a self-programmed set of functions in MATLAB R2012a. The KKT conditions were fed into GAMS with an output file from MATLAB.

5.1. Benchmark and Big M test

As we believe that bilevel problems as discrete scenarios in a multiperiod model is a new combination of frameworks, we performed an initial analysis on the solving this type of mathematical model. Benchmark tests were run for a smaller two stage problem with 8 scenarios. The 8 scenarios were generated by varying the future value weight, biomass pricing, and biomass availability parameters. Several tests were performed using different big M values, and the results are shown in Table 6. While methods have been found to determine the maximum value necessary for the Big M parameter (Vecchiotti et al., 2003), the structure of our problem was different. To solve our bilevel problem, we convert bilinear KKT complementary condition constraints to a pair of Big M integer constraints. While the upper bound of the original constraint can be easily found, the issue lies with the KKT multiplier. Within the context of our problem, there is no upper bound on this variable, meaning a maximum value of M cannot truly be found.

The results show calculation times for solving the two stage problem with enumeration through every biorefinery decision. While calculation time varies little between M values of 300–500, an overall trend of longer calculation time for larger M values can be seen. This indicates the importance of setting the right M value across different problem constraints to ensure reasonable computation time and also optimal solutions. Each set of lower level constraints had M values specific to its grouping. For example, the M values for the set of lumber mill capacity constraints were different than the set for biomass availability. These values were approximated by determining the maximum possible value of the left hand side of the associated KKT constraint and rounding upwards. The optimal solution for the set of M values was compared against a higher reference to ensure consistency. The reference M value was set by steadily increasing the M values until the optimal value became consistent and the solution was reasonable. Values that were too large resulted in longer calculation times and would

Table 7
Optimal values of each biorefinery decision [\$ million].

	New biorefinery	Loc. 1	Loc. 2	Loc. 3
New biorefinery	177.89	–	–	–
Small	–	183.13	189.58	186.08
Medium	–	175.89	179.34	173.91
Large	–	166.48	172.42	162.09

sometimes cause non-discrete values for the binary variables. The non-discrete results are due to the solver converging on values very close to 0 and 1 (Assavapokee, 2004).

M values were adjusted and the problem was solved until the correct solution could no longer be obtained with the reduction of an M value. M values set too low would result in infeasible or inconsistent solutions due to poorly restricted constraints in the KKT conditions. In this problem, M values were set for the lumber mill capacities ($M = 100$), retailer demands ($M = 50$), biomass availability ($M = 200$), and non-negativity constraint for the lower level decision variable ($M = 200$). The reference M was set to 500. Attempts were made to solve the full multistage problem without enumeration; however, calculation times were over 8 h for the varied M values. The solver algorithms may not be able to efficiently handle solving multiple bilevel problems simultaneously. This suggests the need for some form of decomposition algorithm that is less naive than the complete enumeration used here.

5.2. Timberlands system results

With the results of the smaller problem, showing that MILP solvers struggle with bilevel programming scenarios, the full 48 scenario problem was solved by enumerating through all scenarios and all 10 binary decision choices, giving 480 separate problems. Enumeration through the 480 problems required 132.06 s of calculation time. The average time for each problem was 0.275 s, and the times ranged from 0.057 to 1.145 s. Attempts were made to solve problems by enumerating through only the biorefinery decisions, creating 10 problems in total (instead of 480 problems). The large number of scenarios in these problems (48) led to unreasonable calculation times.

The optimal biorefinery decision for our full two stage problem was a small capacity facility at location 2 (Table 7). This decision gave an optimal solution value of \$189.58 million. Overall, the small biorefineries were able to generate a value increase to the system. For medium sized facilities, placement in location 2 was the only option for a more profitable system. Medium biorefineries in locations 1 and 3 as well as all large biorefineries had a lower expected value than a system with no biofuel production.

The expected profits from the harvesters are given in Table 8. The total expected profit yielded by harvesters was \$101.77 million. The expected value from unharvested biomass was \$82.21 million. An average of 2.60 million wet tons of logs were harvested, and an average of 2.15 million wet tons of logs were saved for the future. 54.7% of the available trees were expected to be harvested.

Table 8
Harvester expected profits with small biorefinery at location 2 [\$ million].

Harvester	1	2	3	4	5
Profit	1.03	0.18	1.59	0.04	20.71
Harvester	6	7	8	9	10
Profit	1.23	26.06	6.90	7.89	0.67
Harvester	11	12	13	14	15
Profit	1.59	6.11	26.50	0.77	0.41

Table 9
Harvester expected profits with no biorefinery [\$ million] (harvesters 10–12 inactive).

Harvester	1	2	3	4	5	6
Profit	0.77	0.18	1.66	0.04	21.22	0.44
Harvester	7	8	9	13	14	15
Profit	11.91	6.08	2.08	18.60	0.47	0.15

Table 10
Lumber mill results with small biorefinery in location 2.

Lumber mill	1	2	3	4
Expected profit (\$ million)	42.02	28.80	31.33	36.66
Expected processing capacity (%)	100	100	99.8	100

This means adding the biorefinery did not overly strain the timber supply. Furthermore, almost all of the harvested logs were sent to lumber mills for processing, with only an average of 0.633 million wet tons sold to external markets. This shows that the harvester does not over harvest its materials.

These results can be compared to Table 9, the optimal allocation of a system without a biorefinery. Harvesters 10–12 are inactive. The amount of logs harvested (1.467 million tons) is 43.61% less than the system with the small biorefinery. Without biofuel production, the value of each unit of biomass harvested decreases significantly. A major source of revenue came from converting the low value residual and shavings byproducts to gasoline. This pushed the harvester to provide more materials.

For the optimal biorefinery system, the lumber mill results are shown in Table 10. The lumber mills are extremely profitable and produce at near full capacity. When compared to a system with no biorefinery (Table 11), the profitability of selling byproduct to the biorefinery is shown to push the lumber mills to higher production, yielding an average of 49.7% increase in profit for each mill.

The biorefinery provides an average \$6.48 million profit. Production utilizes an average of 0.78 million dry tons of residuals and 0.591 million dry tons of shavings. The facility runs at an average processing capacity of 78.34%, yielding an average of 130.41 million gallons of gasoline. Analysis of all scenarios determined that the lowest processing capacity for the biorefinery was 67.18%. Even in the worst situations, the biorefinery still produces at a realistic level. The highest capacity reached was 99.93%. From the sales of biomass byproduct, harvesters generate \$25.75 million and manufacturers generate \$5.91 million. By comparison, when sold to the external market, the logging residuals would generate \$6.24 million and the shavings would generate \$4.73 million. Not only is the biorefinery profitable, but it generates additional revenue for the harvesters and manufacturers. Furthermore, the inclusions of gasoline production has no detrimental effects on the established timberlands supply chain. This can be further seen in the lumber mill results in Table 10. The results show that, even with a biorefinery, manufacturers run at full or near full capacity.

To determine the utility of bilevel models as second stage scenarios, a simplified two stage model was compared with a single stage model. For the single stage model, the objective and constraints are a combination of the two levels in the bilevel problem.

Table 11
Lumber mill results with no biorefinery.

Lumber mill	1	2	3	4
Expected profit (\$ million)	28.12	18.12	20.84	25.92
Expected processing capacity (%)	74.4	73.3	73.3	77.0

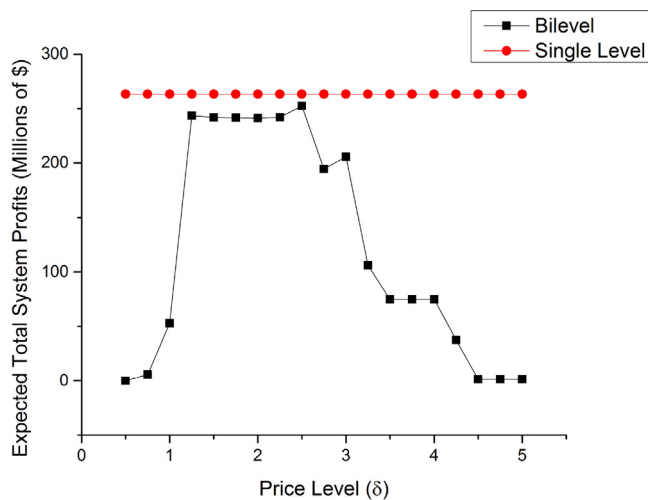


Fig. 2. Varied biomass cost results.

The two stage formulation was simplified by removing the uncertainty in demands and biomass costs, leaving only 8 scenarios in the second stage. These two models were compared by optimizing over a range of biomass costs: $\delta = 0.5$ – 5.0 with intervals of 0.25 . The results are shown in Fig. 2.

While the single level solution showed no variance in behavior with changing biomass cost, the bilevel model's decision making was more complex. The behavior follows similarly to the results from Yeh et al. (2014). With lower biomass costs, harvesters prefer to save material. As prices become more favorable to the harvester, material is provided to the manufacturer for production. As the prices become too high, the biorefinery no longer profits. This system differs in the inclusion of a biorefinery decision and also by the biorefinery being located in the top level. From $\delta = 1.25$ – 3.00 , the optimal solution for the bilevel problem was a small biorefinery in location 2. For all other ranges, the optimal solution was no biorefinery. For the single level problem, the optimal solution was a small biorefinery in location 2. This result shows how bilevel scenarios can differentiate from single level scenarios. It also shows the importance of the multiperiod format; under a deterministic

model, a decision could be made that may be optimal in that specific situation, but not for the range of uncertainties in the system.

It should be noted that these results were specific to the configuration of this supply chain. Given a different set of parameters, it is very possible for another biorefinery decision to be optimal. With a completely different set of uncertain parameters, the solution could vary. For example, if biomass availability is high, a larger biorefinery could yield an optimal solution. These possibilities are more apparent in the next section, where not every scenario's optimal decision was the same as the overall problem's decision.

5.3. Scenario studies

Optimal solutions were also calculated for each scenario. For the results of the 48 scenarios, only 2 decisions were chosen: 11 scenarios were optimal for no biorefinery (NB) while the remaining 37 were optimal for a small facility at location 2. The average value of the optimal solutions was \$192.35 million with a range of \$104.42 million to \$278.60 million. The scenario configurations for the worst and best optimal value solutions are shown in Table 12.

The results of the scenario specific study showed that the minimum processing demand had very little effect on the best value solutions but was important in distinguishing the worst value solution. In the most profitable scenarios, lumber mills already process at full capacity, rendering the demand constraint irrelevant. This means the lower bound on demand is inactive. The results also showed that a higher biomass cost led to a better solution. This result is discussed with Table 13.

The 11 scenarios with optimal solution of no biorefinery were further analyzed. Minimum processing demand t_j had a smaller influence on the biorefinery investment decision. 5 scenarios were 0% demand, 4 were 50% demand, and 2 were 90% demand. The settings of the other parameters were more strict: 9 of the scenarios had higher future value weight (w_{kj}), 9 of the scenarios had lower biomass cost (δ_j), 10 of the scenarios had higher processing costs (μ_j), and 8 scenarios chose availability scenario 1 (v_{ikj} and q_{ikhj}).

Further analysis was performed to determine the effect of the scenario parameters on the optimal scenario solution. The 48 scenarios were compared and grouped. The optimal solutions were grouped based on the scenario parameter being studied. Because each scenario parameter has only 2 values (other than lumber mill demand t_j), the scenario solutions can be paired based on the parameter being analyzed; for any scenario in this set, a partner scenario can be found where the system is the same except for

Table 12
Worst and best value scenarios.

Scenario parameter	Future value weight	Biomass availability	Processing cost multiplier	Biomass cost multiplier	Minimum processing demand
	w_{kj}	q_{ikhj} and v_{ikj}	μ_j	δ_j	t_j
Worst	0.7	Scen. 1	1	1	90%
Best	1.1	Scen. 2	0.7	1.5	Any

Table 13
Scenario parameter impact on optimal solution (scenario 2–scenario 1) [\$ million].

Modifies parameter	Future value	Availability	Biorefinery processing	Biomass cost	Minimum processing demand		
	w_{kj}	q_{ikhj} and v_{ikj}	μ_j	δ_j	t_j		
Scenario 1	0.7	Scen. 1	0.7	1	0%	50%	0%
Scenario 2	1.1	Scen. 2	1	1.5	50%	90%	90%
Average diff.	36.14	67.39	–26.43	37.23	–0.44	–0.44	–0.88
Max. diff.	50.59	57.61	–34.62	35.43	–2.03	–1.73	–3.76
Min. diff.	22.99	75.06	–20.83	39.45	0.00	0.00	0.00
Standard dev.	9.18	7.29	5.94	1.17	0.68	0.63	1.31

Table 14Differences in biomass amounts/facility capacities with changing biomass prices ($[\delta = 1.5] - [\delta = 1.0]$).

	Millions of tons of logs		Processing capacities	
	Harvested	Sold externally	Lumber mills	Biorefinery
Average difference	−0.2626	−0.2633	0.0373%	−4.4882%
Largest deviation	0.3920	−0.9274	1.6153%	−15.9737%

Table 15Differences in profit/value with changing biomass prices ($[\delta = 1.5] - [\delta = 1.0]$) [\$ million].

	Profit		Value	
	Biorefinery	Harvester	Unharvested material	Objective function
Average difference	−11.947	42.862	7.473	38.388

the studied parameter. The differences for these pairs were found for each scenario parameter. For the case of lumber mill demand, the solutions were grouped in sets of 3. The results are shown in Table 13.

It can be seen that the change in t_j had minimal effect on the optimal solution. Overall, biomass availability had the largest impact. Future value and biomass cost were second largest, and processing costs were next. Standard deviation calculations showed that biomass costs and lumber mill demand had the most consistent effects on the optimal solutions. Higher biomass costs led to a higher profit value for the harvester level. While the initial assumption towards this result was due to more revenue being made with higher biomass value, the amount of material harvested is actually lower in these scenarios. Table 14 shows the average changes with changing δ value (the biomass cost multiplier). Despite the lower harvest, the lumber mill production slightly increases (0.0373%), with the largest deviation being 1.6153%. The biorefinery has a large decrease in production due to less residuals being generated. It can also be seen that the decisions to harvest less material deducts more from the external biomass sales than internal.

In the case of the higher biomass costs, less biomass flow creates a more profitable objective. Table 15 shows the differences in profit and value of the upper level problem. The decrease in biorefinery profit and increase in unharvested material total value is consistent with the decrease in material flow, but the harvester profits and overall objective function increase in value. This behavior is consistent with results seen in Yeh et al. (2014). Changing biomass prices creates a situation where a different configuration of active harvesters can yield a better result. This is due to each harvester being unique in amount of available material, harvesting cost, and biomass pricing. Furthermore, higher biomass costs increase the value of each unit sold to the lumber mills. The very slight increase in lumber mill processing shows that the increased biomass prices are not high enough to discourage buying. Therefore, the increase in biomass revenue and unharvested material value could outweigh the losses incurred in the biorefinery from the lower harvests.

5.4. System activity studies

Studies were performed to see how system behavior would change in a situation where certain harvesters and manufacturers were unavailable. The results for inactive harvesters are shown in Table 16. Unharvested material provided no value to the objective function for the harvesters that were inactive. The optimal biorefinery investment decision did not change due to one inactive harvester; no matter which harvester was shut down, a small biorefinery at location 2 was still the optimal choice. One inactive harvester did not have a significant enough effect to discourage

Table 16

Expected value with inactive harvesters [\$ million].

Harvester	1	2	3	4	5
Expected objective value	187.90	189.37	187.85	189.52	167.33
Harvester	6	7	8	9	10
Expected objective value	188.08	141.86	183.52	158.44	177.70
Harvester	11	12	13	14	15
Expected objective value	178.29	180.78	155.74	185.72	188.34

a biorefinery investment. The largest decreases in expected value occurred when zones with large biomass amounts were disabled (e.g. Harvesters 7, 9, and 13); the total unharvested timber value was much larger in these zones. The original optimal objective value was \$189.58 million.

The results for inactive manufacturers are shown in Table 17. A single inactive lumber mill would shift the system to no longer invest in biofuel production. With an inactive manufacturer, a system without a biorefinery is expected to generate much higher profit than one with a smaller biorefinery in location 2 (the optimal decision when biofuel investment is required). These results show that the interaction of the manufacturer on the system has a significant impact on the timberlands system and biofuel production. Demand from manufacturers increases the harvesting activity in the system. Also, the additional supply of shavings byproduct assists in the economic viability of the biorefinery.

Two sets of serial deletion studies were run across the set of harvesters. Results for these studies are shown in Appendix B. The harvesters were sorted by impact on the estimated value. The first study deactivated harvesters in the order of smallest impact (Table B.1). This study determined that the system can deactivate up to 13 harvesters and still function. Deactivating another harvester pushes the system into an infeasible region; the 90%

Table 17

Expected value with inactive lumber mills [\$ million].

Inactive lumber mill	1	2	3	4
Optimal value w/o biorefinery	173.057	176.481	175.267	172.054
Optimal value w/ biorefinery	146.889	167.828	159.388	145.535
Optimal biorefinery decision	Small loc. 2	Small loc. 2	Small loc. 2	Small loc. 2

processing capacity constraints on the manufacturer could no longer be satisfied. Removing harvesters showed an overall trend in decreasing difference between the optimal estimated value and the estimated value of a system without a biorefinery. These two values never converge due to reaching problem infeasibility.

In a second serial deletion study, harvesters were deactivated starting from the largest impact on the expected value (B.2). Whenever an infeasibility was encountered, that harvester was reactivated and the harvester with the next largest impact was deactivated. These results also showed a decreasing trend in difference between the optimal estimated value and the estimated value with no biorefinery. The smallest difference occurred when the last possible harvester was deactivated with a value of \$2.98 million.

6. Conclusions

This study explored the development of a two stage multi-period bilevel model to represent biorefinery investments in an established timberlands system under uncertainty. The calculation times and result validity were reliant on the correct selection of big M values. Enumerating the biorefinery decisions through all scenarios was shown to be much faster than solving the whole two stage problem using the standard branch-and-bound algorithm. Applying the optimization model to a particular case study demonstrated that such a model can provide a useful initial analysis of timber management implementing biofuels into its supply chain. The results showed that 1 biorefinery could coexist with the established lumber mills without detrimental effect on the original supply flows. The optimal solution showed that the biorefinery could be profitable as well as provide additional revenue to the system through byproduct purchases. It was shown that the bilevel scenarios and multiperiod format provided a more complete view of the model compared to the more limited scope of single level and deterministic models. Further analysis was performed to determine the influence of the scenario parameters on the system behavior. Finally, system activity studies were performed by deactivating harvesters and lumber mills. It was discovered that the biorefinery was very dependent on the byproduct generated from all 4 lumber mills. Serial deletion studies showed that the system could still operate with 13 out of 15 harvesters deactivated. Also, no matter which harvesters were inactive, a small biorefinery at location 2 was always a more profitable

choice than no biorefinery. Further developments of this model include increasing the supply chain size, which would allow for investment of more than 1 biorefinery facility. The increase in size and complexity would lead to a more difficult problem to solve. Alternative solution methods may be required to accommodate for these improvements. We believe that this research has led to a useful decision analysis tool for timberlands management that are considering the development of biofuels.

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Appendix A. Simulation data

See [Tables A.1–A.10](#).

Table A.1
Timberlands system index values.

Index	Range	Value
i	1–15	Harvest zones
m	1–4	Lumber mills
k	1 2 3	Logs Logging residuals Shavings
l	1 2	Lumber Gasoline
h	1 2	Preallocated level Extra level
u	Small, medium, large	Biorefinery capacity
e	Loc. 1, Loc. 2, Loc. 3	Biorefinery location
j	1–48	Scenario

Table A.2
Biomass selling prices[logs:/wet ton, residuals:/dry ton]

Harvester ($i=$)		1	2	3	4	5
Internal price (c_{ik})	Logs	\$48.47	\$50.90	\$50.90	\$44.70	\$44.70
	Residuals	\$27.50	\$27.50	\$27.50	\$27.50	\$25.00
External price (o_{ik})	Logs	\$38.78	\$40.72	\$40.72	\$35.76	\$35.76
	Residuals	\$8.00	\$8.00	\$8.00	\$8.00	\$8.00
Harvester ($i=$)		6	7	8	9	10
Internal price (c_{ik})	Logs	\$44.70	\$38.68	\$38.68	\$38.68	\$29.33
	Residuals	\$25.00	\$25.00	\$30.00	\$30.00	\$27.50
External price (o_{ik})	Logs	\$35.76	\$30.94	\$30.94	\$30.94	\$23.46
	Residuals	\$8.00	\$8.00	\$8.00	\$8.00	\$8.00
Harvester ($i=$)		11	12	13	14	15
Internal price (c_{ik})	Logs	\$29.33	\$37.85	\$37.85	\$37.85	\$37.85
	Residuals	\$27.50	\$27.50	\$27.50	\$27.50	\$30.00
External price (o_{ik})	Logs	\$23.46	\$30.28	\$30.28	\$30.28	\$30.28
	Residuals	\$8.00	\$8.00	\$8.00	\$8.00	\$8.00

Table A.3Harvesting costs ($k = \text{logs}$, $h = 2$).

Harvester ($i =$)	1	2	3	4	5
Processing costs (r_{ikh}) [/wet ton]	\$9.60	\$10.09	\$10.09	\$8.82	\$8.82
Fixed costs (f_{ik})	\$106.00	\$6.41	\$73.30	\$1.61	\$663.80
Harvester ($i =$)	6	7	8	9	10
Processing costs (r_{ikh}) [/wet ton]	\$8.82	\$7.62	\$7.62	\$7.62	\$5.44
Fixed costs (f_{ik})	\$63.97	\$2,417.00	\$35.16	\$1,618.06	\$355.80
Harvester ($i =$)	11	12	13	14	15
Processing costs (r_{ikh}) [/wet ton]	\$5.44	\$7.42	\$7.42	\$7.42	\$7.42
Fixed costs (f_{ik})	\$350.29	\$226.87	\$922.14	\$159.08	\$37.81

Table A.4Yield (α_{kl}) [logs:wet tons/others:dry tons].

	Lumber [ton/ton]	Gasoline [gallons/ton]
Logs	0.27	47.75
Residuals	–	92.21
Shavings	–	98.80

Table A.5Transportation costs (s_{im}) [/ton].

Harvester (i)	Lumber mills (m)			
	1	2	3	4
1	\$4.20	\$4.81	\$7.23	\$9.62
2	\$6.84	\$5.82	\$9.62	\$7.68
3	\$8.11	\$5.82	\$9.34	\$6.17
4	\$8.75	\$6.84	\$7.47	\$5.47
5	\$7.16	\$5.13	\$9.39	\$4.37
6	\$5.82	\$3.62	\$5.47	\$5.47
7	\$2.61	\$4.45	\$4.45	\$8.12
8	\$4.60	\$6.50	\$3.50	\$9.00
9	\$4.51	\$5.89	\$3.26	\$7.68
10	\$7.54	\$7.54	\$3.90	\$8.23
11	\$12.75	\$6.38	\$4.60	\$4.60
12	\$7.79	\$4.75	\$4.37	\$4.37
13	\$9.39	\$6.52	\$5.82	\$3.62
14	\$14.00	\$9.07	\$7.47	\$5.82
15	\$13.50	\$9.07	\$7.79	\$5.47

Table A.6Lumber mill parameters ($k = \text{logs}$, $l = \text{lumber}$).

		Shavings price [\$/dry ton]	Capacity [10 ⁶ wet tons]	100% process capacity [10 ⁴ product tons]	Lumber price [\$/product ton]	Processing cost [\$/ton]
Units						
Parameter		$\hat{c}_{mk}$	G_m	A_{ml}	p_{ml}	n_m
Lumber mills (m)	1	\$10.00	0.53	14.31	\$588	\$29
	2	\$10.00	0.52	14.04	\$504	\$28
	3	\$10.00	0.46	12.42	\$550	\$32
	4	\$10.00	0.46	12.42	\$588	\$29

Table A.9Biorefinery parameters ($l = \text{gasoline}$).

Facility size	Capacity	Product price $\hat{p}_{lue}$ [/gallon]			Processing cost	Annualized capital cost β_{lue} [\$ million]		
Index u	$\hat{G}_u$ [tons]	Loc. 1	Loc. 2	Loc. 3	$\hat{n}_u$ [/ton]	Loc. 1	Loc. 2	Loc. 3
Small	175	\$1.454	\$1.485	\$1.488	\$73.20	\$61.737	\$59.885	\$64.207
Medium	250	\$1.458	\$1.464	\$1.448	\$67.00	\$77.448	\$75.877	\$79.090
Large	350	\$1.424	\$1.442	\$1.403	\$57.77	\$94.274	\$92.389	\$96.160

Table A.7Transportation costs from harvester to biorefinery ($\hat{s}_{ie}$) [/ton of biomass].

Harvesters	Loc. 1	Loc. 2	Loc. 3
1	\$9.52	\$10.82	\$10.11
2	\$7.47	\$7.48	\$10.66
3	\$6.96	\$5.67	\$11.45
4	\$7.31	\$8.21	\$6.77
5	\$7.88	\$8.82	\$4.75
6	\$3.45	\$4.32	\$6.23
7	\$5.37	\$8.13	\$4.64
8	\$9.55	\$10.58	\$12.95
9	\$8.28	\$6.27	\$13.58
10	\$9.71	\$6.27	\$8.99
11	\$5.04	\$2.35	\$8.74
12	\$8.58	\$6.88	\$7.22
13	\$7.48	\$9.15	\$12.16
14	\$11.77	\$13.03	\$5.37
15	\$7.48	\$10.19	\$8.61

Table A.8Cost of acquisition for biorefinery from facilities: biomass price and transportation cost combined (g_{mke}) [/ton of biomass].

Lumber mill	Loc. 1	Loc. 2	Loc. 3
1	\$22.75	\$18.13	\$15.04
2	\$23.25	\$20.58	\$18.58
3	\$19.52	\$16.27	\$17.48
4	\$17.47	\$16.27	\$21.77

Table A.10Biomass availability scenarios (v_{ikj}) [10^4 wet tons of logs].

Harvester	Scenario	
	1	2
1	4.86	3.43
2	0.28	0.52
3	3.20	4.47
4	0.08	0.14
5	33.00	67.17
6	3.18	4.05
7	138.86	122.52
8	2.02	24.78
9	92.96	77.48
10	26.96	39.77
11	26.54	36.70
12	13.32	31.46
13	54.14	111.66
14	9.34	11.54
15	2.22	4.22

Appendix B. Additional results

The inactive harvester series column details the order of deactivation; moving down a row deactivates an additional harvester (Tables B.1 and B.2).

Table B.1

Serial deletion: smallest impact series.

Inactive harvester series	Expected value [\$ million]									
	NB	S1	S2	S3	M1	M2	M3	L1	L2	L3
4	177.85	183.08	189.52	186.03	175.84	179.28	173.86	166.43	172.36	162.04
2	177.66	182.88	189.32	185.81	175.63	179.08	173.64	166.22	172.15	161.82
15	176.51	181.62	188.09	184.53	174.35	177.81	172.35	164.90	170.83	160.50
6	175.11	180.00	186.59	182.96	172.74	176.28	170.78	163.27	169.28	158.91
1	173.47	178.43	184.98	181.41	171.15	174.67	169.22	161.67	167.64	157.34
3	171.91	176.71	183.23	179.68	169.41	172.88	167.48	159.89	165.80	155.61
14	168.10	172.53	179.40	175.23	165.17	168.92	162.93	155.55	161.71	151.07
8	162.55	166.36	173.12	169.22	158.79	162.48	156.75	149.01	155.07	144.76
12	153.49	156.67	163.26	159.23	148.75	152.27	146.55	138.90	144.86	134.51
11	142.25	145.04	151.60	147.77	136.96	140.30	135.06	126.74	132.24	122.92
10	130.32	132.27	138.56	134.69	124.04	126.93	121.86	113.59	118.43	109.57
5	107.63	109.03	115.30	110.73	100.52	103.55	97.87	89.81	94.75	85.43
9	76.35	77.13	82.67	79.24	68.19	70.72	66.30	57.11	61.33	53.58

Table B.2

Serial deletion: largest impact series.

Harvesters		Expected value [\$ million]									
Added	Removed	NB	S1	S2	S3	M1	M2	M3	L1	L2	L3
–	7	131.30	134.71	141.86	136.35	126.47	130.60	123.65	116.11	122.58	111.30
–	13	99.75	99.08	106.87	101.41	90.30	95.14	88.48	79.57	86.53	75.93
–	9	–	–	–	–	Infeasible	–	–	–	–	–
9	5	76.07	72.73	80.56	73.95	63.58	68.55	60.92	52.45	59.18	48.12
–	10	–	–	–	–	Infeasible	–	–	–	–	–
10	11	–	–	–	–	Infeasible	–	–	–	–	–
11	12	–	–	–	–	Infeasible	–	–	–	–	–
12	8	69.22	65.07	73.05	66.34	55.88	60.94	53.32	44.58	51.35	40.44
–	14	–	–	–	–	Infeasible	–	–	–	–	–
14	3	67.25	62.71	70.63	64.00	53.44	58.46	50.95	42.07	48.78	38.02
–	1	–	–	–	–	Infeasible	–	–	–	–	–
1	6	–	–	–	–	Infeasible	–	–	–	–	–
6	15	65.98	61.08	68.97	62.32	51.75	56.78	49.27	40.33	46.99	36.30
–	2	–	–	–	–	Infeasible	–	–	–	–	–
2	4	65.93	61.02	68.91	62.26	51.69	56.71	49.21	40.26	46.93	36.23

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